

A CANONICAL BOX PARTITION OF THE SET OF G -PARKING FUNCTIONS, WITH AN APPLICATION TO THE SPANNING TREES OF THE HYPERCUBE

WORKING NOTE — QNPARKINGFUNCTIONS PROJECT

ABSTRACT. For a connected multigraph G with root q and a total order on the vertices, we show that the fibers of the map sending each G -parking function to its lexicographically minimal Dhar burn order are coordinate boxes. The boxes partition the set of all G -parking functions; their tops are exactly the maximal G -parking functions. This yields sign-free formulas for $T_G(1, 1)$ and $T_G(1, y)$ as sums of products of q -integers over acyclic orientations with unique sink, gives an explicit and canonical construction of an M -partition of the parking-function multicomplex whose existence follows from Merino's inductive proof of Chari's conjecture for cographic matroids, and, for the hypercube Q_n , contains the canonical box of Benson–Chakrabarty–Tetali as its largest cell. We also prove a support-product theorem for the spanning trees of $G \times K_2$ whose iteration telescopes to Stanley's formula for the number of spanning trees of Q_n , give an explicit bijection realizing its full-support slice, and state precisely the bijective step that remains open.

1. SETUP

Let $G = (V, E)$ be a connected multigraph with n vertices, a distinguished root q , and a fixed total order on V with q smallest; we write $V = \{v : 0 \leq v \leq n - 1\}$ with $q = 0$. For $u, v \in V$ let $a(u, v)$ denote the number of edges between u and v .

A function $f : V \setminus \{q\} \rightarrow \mathbb{Z}_{\geq 0}$ is a *G -parking function* (equivalently, a *superstable configuration*) if for every nonempty $A \subseteq V \setminus \{q\}$ there is $v \in A$ with $f(v) < \sum_{u \notin A} a(v, u)$. We write $\mathcal{P}(G, q)$ for the set of G -parking functions and recall Dhar's criterion [7, 2]: $f \in \mathcal{P}(G, q)$ if and only if the following process burns every vertex. Initially only q is burnt; a vertex v may burn whenever

$$f(v) < w(v) := \sum_{u \text{ burnt}} a(v, u),$$

and the process is abelian: if some order burns everything, every maximal sequence of legal burns does.

By downward closure ([2, Prop. 2.2]) $\mathcal{P}(G, q)$ is an order ideal of $\mathbb{Z}_{\geq 0}^{V \setminus \{q\}}$ under the coordinatewise order; all its maximal elements have the same total degree $\bar{g} := |E| - |V| + 1$ ([2, Cor. 3.2]), they are in bijection with the acyclic orientations of G with unique source at q ([2, Thm. 3.1]), and their number is the Tutte evaluation $T_G(1, 0)$ ([2, Thm. 4.1]).

Definition 1.1. For $f \in \mathcal{P}(G, q)$, the *canonical burn order* $\sigma(f)$ is the sequence obtained by always burning the *smallest* burnable vertex. By the abelian property $\sigma(f)$ is a well-defined ordering $(\sigma_0 = q, \sigma_1, \dots, \sigma_{n-1})$ of V .

For an arbitrary ordering σ of V starting at q and $0 \leq k < n$ write $w_k(v) = \sum_{j < k} a(v, \sigma_j)$ for the burnt-neighbor weight of v before step k . Define, for $v = \sigma_k$,

$$\text{top}_\sigma(v) = w_k(v) - 1, \quad \text{bot}_\sigma(v) = \max(\{0\} \cup \{w_j(v) : j < k, \sigma_j > v\}).$$

Since $w_\bullet(v)$ is nondecreasing, $\text{bot}_\sigma(v)$ is the burnt-neighbor weight of v at the *last* step that burned a larger vertex while v was unburnt (its “skip watermark”), and 0 if there was no such step.

2. THE FIBER THEOREM

Theorem 2.1. *Let $f \in \mathcal{P}(G, q)$ and $\sigma = \sigma(f)$. Then*

$$\{g \in \mathcal{P}(G, q) : \sigma(g) = \sigma\} = \{g : \text{bot}_\sigma \leq g \leq \text{top}_\sigma \text{ coordinatewise}\},$$

and moreover every integer vector in the right-hand box is a G -parking function.

Proof. ($\subseteq$) Let $g \in \mathcal{P}(G, q)$ with $\sigma(g) = \sigma$ and let $v = \sigma_k$. Since v burns at step k we have $g(v) < w_k(v)$, i.e. $g(v) \leq \text{top}_\sigma(v)$. If $j < k$ and $\sigma_j > v$, then at step j the canonical process preferred the larger vertex σ_j , so v was not burnable: $g(v) \geq w_j(v)$. Taking the maximum over such j gives $g(v) \geq \text{bot}_\sigma(v)$.

($\supseteq$) Let g be any integer vector with $\text{bot}_\sigma \leq g \leq \text{top}_\sigma$. We show by induction on k that the canonical process for g burns $\sigma_1, \dots, \sigma_k$ in this order. Assume it has burnt exactly $\{\sigma_0, \dots, \sigma_{k-1}\}$, so each unburnt v has burnt-neighbor weight $w_k(v)$. First, $v = \sigma_k$ is burnable: $g(v) \leq \text{top}_\sigma(v) = w_k(v) - 1 < w_k(v)$. Second, let $u < \sigma_k$ be unburnt; then $u = \sigma_m$ for some $m > k$, so u is skipped at step k in σ , whence $\text{bot}_\sigma(u) \geq w_k(u)$ and therefore $g(u) \geq w_k(u)$: u is not burnable. Hence the smallest burnable vertex is exactly σ_k . By induction the process burns all of V , so $g \in \mathcal{P}(G, q)$ and $\sigma(g) = \sigma$. $\square$

Corollary 2.2. *The boxes $B_\sigma := [\text{bot}_\sigma, \text{top}_\sigma]$, over the canonical burn orders σ of parking functions, partition $\mathcal{P}(G, q)$. The map $\sigma \mapsto \text{top}_\sigma$ is a bijection onto the set of maximal G -parking functions; in particular the number of boxes is $T_G(1, 0)$.*

Proof. The fibers of $f \mapsto \sigma(f)$ partition $\mathcal{P}(G, q)$, and by Theorem 2.1 each fiber is a box. Every edge of G contributes its multiplicity to $w_k(\sigma_k)$ exactly once, at the step burning its later endpoint, so $\sum_v \text{top}_\sigma(v) = |E| - (n - 1) = g$; hence each top is a maximal parking function [2, Prop. 2.3, Cor. 3.2]. The top lies in its own box, so $\sigma(\text{top}_\sigma) = \sigma$ and $\sigma \mapsto \text{top}_\sigma$ is injective. If m is maximal then $m \in B_{\sigma(m)}$, so $m \leq \text{top}_{\sigma(m)}$; both have degree g , forcing $m = \text{top}_{\sigma(m)}$. The count is [2, Thm. 4.1]. $\square$

Corollary 2.3 (sign-free Tutte formulas). *Writing $[k]_y = 1 + y + \dots + y^{k-1}$ and $b_m := \text{bot}_{\sigma(m)}$ for a maximal parking function m ,*

$$T_G(1, y) = \sum_{m \text{ maximal}} \prod_{v \neq q} [m(v) - b_m(v) + 1]_y, \quad T_G(1, 1) = \sum_m \prod_{v \neq q} (m(v) - b_m(v) + 1).$$

Proof. By Merino’s theorem [9], $T_G(1, y) = \sum_{f \in \mathcal{P}(G, q)} y^{g - \deg f}$. Group the sum by the boxes of Corollary 2.2: within B_σ , the complement $\text{top}_\sigma(v) - f(v)$ ranges independently over $\{0, \dots, \text{top}_\sigma(v) - \text{bot}_\sigma(v)\}$. $\square$

Via [2, Thm. 3.1] the index set is the set of acyclic orientations with unique source at q , and $m(v) + 1$ is the weighted in-degree of v ; the watermark b_m is a statistic on such orientations that appears to be new. Corollary 2.3 replaces the 2^k -term inclusion–exclusion of [2, Cor. 3.3] with a sum of $T_G(1, 0)$ positive terms.

Proposition 2.4 (the identity cell). *Suppose the vertex labeling is such that every prefix $\{0, 1, \dots, k\}$ induces a connected subgraph. Then the identity ordering $\iota = (0, 1, \dots, n-1)$ is a canonical burn order, $\text{bot}_\iota \equiv 0$, and $\text{top}_\iota(v) = (\sum_{u < v} a(v, u)) - 1$. For the hypercube Q_n with its binary labeling, $\text{top}_\iota(v) = \text{wt}(v) - 1$ is the canonical parking function of [2, Thm. 5.2] and the cell B_ι is its entire lower box, of cardinality $\prod_{v \neq 0} \text{wt}(v) = \prod_{k=2}^n k^{\binom{n}{k}}$.*

Proof. Burning in increasing label order never skips a smaller unburnt vertex, so no watermark is ever raised and each step is legal because v has a burnt (smaller) neighbor by the prefix-connectivity assumption. In Q_n the smaller neighbors of v are exactly the $\text{wt}(v)$ vertices obtained by clearing one bit. $\square$

Conjecture 2.5 (M -shelling). *In the lexicographic order of the canonical burn orders, every initial union of boxes of Corollary 2.2 is an order ideal; that is, the partition together with this order is an M -shelling in the sense of Chari [4].*

Conjecture 2.5 has been verified exhaustively for Q_2 , Q_3 , a 4-vertex multigraph, and 25 pseudorandom connected multigraphs on up to 6 vertices. The *existence* of an M -shelling of this multicomplex is a theorem of Merino [10, Thm. 5.8], proved by induction on parallel classes; the construction above is explicit and appears to be the first closed-form one (a sweep of the citation trails of [10, 4], of the Gröbner-theoretic parking-function literature, and of the sandpile literature through 2026 found no interval partition of $\mathcal{P}(G, q)$ for general G). Two pieces of prior art deserve emphasis: for complete graphs, the fibers of the outcome map of classical parking functions are known to be boxes with a product formula [5, Prop. 3.3] — Theorem 2.1 specializes to this; and the greedy-burning engine underlying $\sigma(f)$ is the same one Cori and Le Borgne used for their activity-preserving bijection to spanning trees [6], of which the present partition can be regarded as the fiber-level strengthening. On the algorithmic side, the search tree of Theorem 2.1 enumerates the maximal parking functions of G (equivalently, by [2, Thm. 3.1], the acyclic orientations with unique fixed source) one per leaf with no duplicate suppression; we are not aware of a published enumeration algorithm stated for maximal parking functions.

3. THE SUPPORT-PRODUCT CONJECTURE FOR Q_n

Write $Q_n = Q_{n-1} \times K_2$ and fix the direction $d = n$ (all directions are equivalent). A spanning tree T of Q_n contains, for each $u \in V(Q_{n-1})$, at most one *vertical* edge $\{(u, 0), (u, 1)\}$; let $S(T) \subseteq V(Q_{n-1})$ be the set of u for which it does. For $S \subseteq V(Q_{n-1})$ let $F_{Q_{n-1}}(S)$ denote the number of spanning forests of Q_{n-1} with exactly $|S|$ components, one containing each vertex of S (equivalently, by the all-minors Matrix-Tree theorem, the determinant of the Laplacian of Q_{n-1} with the rows and columns of S deleted).

Theorem 3.1 (support product). *Let G be any connected graph with M vertices and $c(G)$ spanning trees. For every nonempty $S \subseteq V(G)$, the number of spanning trees of $G \times K_2$ with vertical support exactly S is $\frac{1}{2} c(G) 2^{|S|} F_G(S)$. In particular, for $G = Q_{n-1}$ the number of pairs (T, r) with T a spanning tree of Q_n , r a root, and $S(T) = S$ equals*

$$N(S) = 2^{|S|} \cdot F_{Q_{n-1}}(S) \cdot 2^{n-1} c(Q_{n-1}).$$

Proof. Give the vertical edge at u the weight z_u and every horizontal edge weight 1, and let $t(z) = \sum_T \prod_{u \in S(T)} z_u$ be the weighted spanning-tree sum of $G \times K_2$. Order the vertices as two level-blocks; the weighted Laplacian is $\begin{pmatrix} L_G + Z & -Z \\ -Z & L_G + Z \end{pmatrix}$ with $Z = \text{diag}(z_u)$. Conjugating by the orthogonal level-swap basis $\frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ block-diagonalizes it as $L_G \oplus (L_G + 2Z)$. The

product of the nonzero Laplacian eigenvalues of a weighted graph on N vertices equals N times its tree sum; the zero eigenvalue lives in the L_G block, so with $N = 2M$,

$$t(z) = \frac{1}{2M} \left(\prod_{\lambda \neq 0} \lambda(L_G) \right) \det(L_G + 2Z) = \frac{c(G)}{2} \det(L_G + 2Z).$$

Expanding the determinant by principal minors in the diagonal perturbation $2Z$ and applying the all-minors Matrix-Tree theorem, $\det(L_G + 2Z) = \sum_R 2^{|R|} F_G(R) \prod_{u \in R} z_u$. Both sides of $t(z)$ are multilinear in the z_u , and comparing the coefficient of $\prod_{u \in S} z_u$ gives the claim; multiplying by the $2M = 2^n$ root choices gives the hypercube form. $\square$

Proposition 3.2 (full-support slice, bijectively). *For $S = V(G)$ the theorem has an explicit bijection: a spanning tree of $G \times K_2$ containing every vertical edge is exactly a choice of a spanning tree of G (its image under contracting all vertical edges) together with a free level in $\{0, 1\}$ for each of its $M - 1$ edges; hence there are $c(G) 2^{M-1} = \frac{1}{2} c(G) 2^{|S|} F_G(S)$ of them, as $F_G(V(G)) = 1$.*

Remark 3.3 (partial doubles). For general S , contracting the vertical edges of a tree with support S gives a bijection with the spanning trees of the *partial double* $G^{(S)}$: the graph obtained from G by splitting every vertex $u \notin S$ into two copies u_0, u_1 (an edge (u, v) becomes two parallel edges if $u, v \in S$; the pair $(u, v_0), (u, v_1)$ if only $u \in S$; and $(u_0, v_0), (u_1, v_1)$ if $u, v \notin S$). Theorem 3.1 is thus equivalent to the identity $c(G^{(S)}) = \frac{1}{2} c(G) 2^{|S|} F_G(S)$, and the remaining *bijective* problem is to realize this identity by an explicit map — for instance by splitting one vertex at a time.

Corollary 3.4. *Stanley's formula $c(Q_n) = 2^{2^n - n - 1} \prod_{k=1}^n k \binom{n}{k}$ follows by iterating Theorem 3.1 over the n directions.*

Proof. Summing over S and using $\sum_S z^{|S|} F_G(S) = \det(zI + L_G)$ (all-minors Matrix-Tree; the $S = \emptyset$ term vanishes) at $z = 2$,

$$2^n c(Q_n) = \sum_{S \neq \emptyset} N(S) = 2^{n-1} c(Q_{n-1}) \cdot \det(2I + L_{Q_{n-1}}) = 2^{n-1} c(Q_{n-1}) \prod_{k=0}^{n-1} (2k+2) \binom{n-1}{k}.$$

Writing $R_n := 2^n c(Q_n)$, this is $R_n = R_{n-1} \prod_{j=1}^n (2j) \binom{n-1}{j-1}$, and by induction with Pascal's rule $\binom{n-1}{j} + \binom{n-1}{j-1} = \binom{n}{j}$ one obtains $R_n = \prod_{j=1}^n (2j) \binom{n}{j}$, which is Stanley's formula. $\square$

Theorem 3.1 was found experimentally and confirmed *exactly, class by class*, for $n = 2$, for $n = 3$ (all 15 supports), and for $n = 4$ (all 255 supports, from the complete set of 679 477 248 spanning-tree-root pairs) before the proof above was identified; the identity is elementary by weighted Matrix-Tree arguments, and closely related product-graph computations appear in [3] and in Martin-Reiner, so we regard the *statement* as possibly folklore — the content for this project is its bijective realization. Two supporting facts, verified computationally: (i) conditioned on S , the spins of the vertical edges (the side of each vertical edge nearer the root) are *exactly* jointly uniform over $\{0, 1\}^S$ for the ensemble of (tree, root) pairs — this is a transported and marginalized form of Bernardi's independence theorem [3, Thm. 1], and accounts for the factor $2^{|S|}$; (ii) the uniformity fails for trees with a fixed root, so any bijective realization must work with rooted-anywhere trees and only afterwards quotient by the 2^n root choices.

A bijective proof of Theorem 3.1 would amount to a bijection

$$\{(T, r) : S(T) = S\} \longleftrightarrow \{0, 1\}^S \times \mathcal{F}_{Q_{n-1}}(S) \times \{\text{rooted spanning trees of } Q_{n-1}\},$$

and, by Proposition 3.4, its iteration over n would constitute a fully bijective proof of Stanley’s formula — the problem stated as open in [8, Example 5.6.10] and, for the bijective form, still open after [3] (cf. the Notes to Chapter 5 of the second edition of [8]).

A structural reformulation: deleting the vertical edges of T leaves forests H_0, H_1 in the two levels, and T is a tree if and only if the bipartite “component graph” on the components of H_0 and H_1 , with one edge per element of S , is a tree. Conjecture 3.1 asserts that these glued structures are equinumerous with the product above; finding the gluing bijection is the core remaining problem.

4. COMPUTATIONAL APPENDIX

All computations were performed with `qpark` (Rust, in this repository): exact enumeration of maximal parking functions via canonical burn orders with the skip-watermark pruning of Theorem 2.1 (Q_4 : 3040575 maximal parking functions and 42467328 spanning trees in 1.5s, matching the chromatic polynomial of Q_4 , the Matrix-Tree determinant, and Stanley’s formula); the degree enumerator $T_{Q_4}(1, y)$; Smith normal forms of the reduced Laplacians, giving $K(Q_5) = \mathbb{Z}_2^5 \times \mathbb{Z}_6 \times \mathbb{Z}_{24}^4 \times \mathbb{Z}_{48} \times \mathbb{Z}_{192}^3 \times \mathbb{Z}_{960}$ in agreement with all three theorems of Bai [1]; and the census and spin experiments described above. Validation is by exhaustive comparison against brute-force Dhar enumeration on small and random multigraphs.

REFERENCES

- [1] H. Bai, *On the critical group of the n -cube*, Linear Algebra Appl. 369 (2003) 251–261.
- [2] B. Benson, D. Chakrabarty, P. Tetali, *G -parking functions, acyclic orientations and spanning trees*, Discrete Math. 310 (2010) 1340–1353.
- [3] O. Bernardi, *On the spanning trees of the hypercube and other products of graphs*, Electron. J. Combin. 19(4) (2012), #P51.
- [4] M. K. Chari, *Two decompositions in topological combinatorics with applications to matroid complexes*, Trans. Amer. Math. Soc. 349 (1997) 3925–3943.
- [5] L. Colmenarejo, P. E. Harris, Z. Jones, C. Keller, A. Ramos Rodríguez, E. Sukarto, A. R. Vindas-Meléndez, *Counting k -Naples parking functions through permutations and the k -Naples area statistic*, Enumer. Comb. Appl. 1(2) (2021), S2R11.
- [6] R. Cori, Y. Le Borgne, *The sand-pile model and Tutte polynomials*, Adv. Appl. Math. 30 (2003) 44–52.
- [7] D. Dhar, *Self-organized critical state of sandpile automaton models*, Phys. Rev. Lett. 64 (1990) 1613–1616.
- [8] R. P. Stanley, *Enumerative Combinatorics, Volume 2*, Cambridge Univ. Press, 1999; 2nd ed. 2024, Example 5.6.10.
- [9] C. Merino López, *Chip firing and the Tutte polynomial*, Ann. Comb. 1 (1997) 253–259.
- [10] C. Merino, *The chip firing game and matroid complexes*, Discrete Math. Theor. Comput. Sci. Proc. AA (2001) 245–256.